



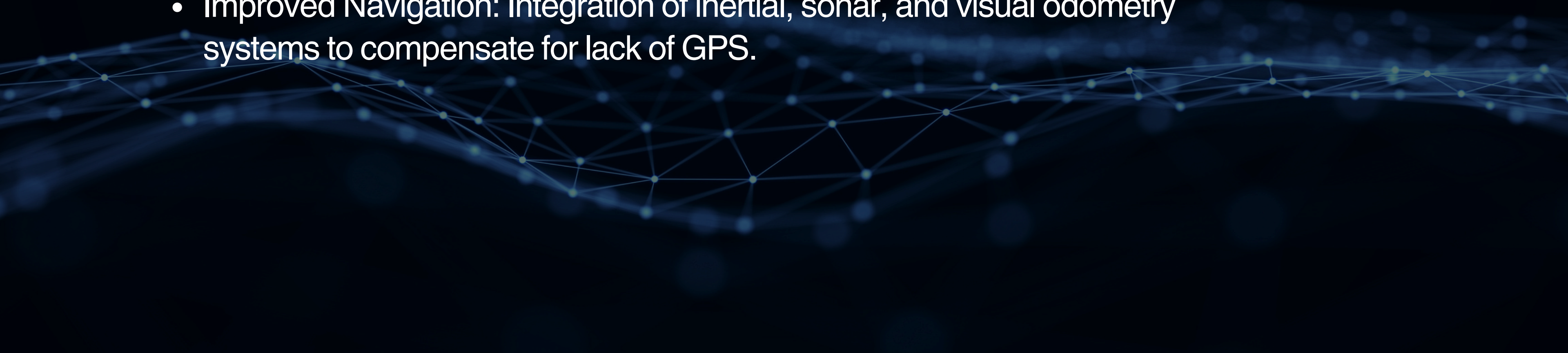
UNDERWATER ROBOTICS CHALLENGES

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FUTURE CAPABILITIES



- Extended Autonomy: Better on-board decision-making using AI and machine learning.
 - Collaborative Swarms: Groups of AUVs working together for search, mapping, or inspection tasks.
 - Improved Navigation: Integration of inertial, sonar, and visual odometry systems to compensate for lack of GPS.
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ETHICS



- Military Dual-Use: Robots used for surveillance and mine detection may also be weaponized.
- Environmental Surveillance: Raises concerns about continuous monitoring of protected marine zones.
- Data Privacy: Sensitive marine research data may have geopolitical implications.

SUSTAINABILITY ISSUES



- Acoustic Pollution: Sonar and thrusters contribute to underwater noise pollution, disturbing marine life.
 - Eco-Friendly Materials: Need for biodegradable or recyclable materials to avoid contributing to ocean waste.
 - Marine Habitat Disruption: Robots may damage delicate coral reefs or stir up sediments.
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THANK YOU

